

Advanced Python: Applied AI and Machine Learning Techniques

Duration: 8 modules (8 days - half-day sessions)

Course Description:

This **8-day intensive training program** is designed for professionals looking to master advanced Python techniques and applied machine learning (ML) skills. The course provides hands-on experience with cutting-edge tools and frameworks, including **Jupyter Notebook, NumPy, Pandas, PyTorch, Hugging Face Transformers**, and more. Participants will explore a wide range of topics, from **data manipulation and deep learning** to **image recognition, generation, and transformation techniques**, as well as **state-of-the-art natural language processing (NLP)**.

Target Audience (who should attend):

- Data scientists, ML engineers, and software developers looking to advance their Python and ML skills.
 - Professionals interested in **image processing, NLP, and generative AI**.
 - Teams aiming to integrate **state-of-the-art ML techniques** into their workflows.
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Course Track: Intermediate/Advanced

Pre-requisites:

- Participants are expected to know **Core Python** concepts that includes the following:
 - Python statements and syntax
 - Flow control statements
 - Defining functions and function prototypes
 - Built-in data-types (numbers, strings, tuple, list, set, dict)
 - Loading modules and using standard libraries bundled with python (`sys`, `time`, `random`, `string`)
 - Python OOP concepts

Recommended training preparation:

- Participants attending this training are expected to learn basic python concepts by reading and practising concepts covered within Python tutorial available at the URL below:
 - <https://docs.python.org/3/tutorial/index.html>
- Participants are also expected to learn from **Chapter 1 to Chapter 6.1** from the Python standard library documentation available at the URL below:
 - <https://docs.python.org/3/library/index.html>

Course Objectives:

- Explore a variety of tools and frameworks available in Python for Data Analytics, AI and ML
- Learn how to setup and use Anaconda Python with Jupyter

- Learn how to preprocess and analyze data using **NumPy** and **Pandas**
- Build and train **deep learning models** with **PyTorch**, including Convolutional Neural Networks (CNNs) for image recognition and Generative Adversarial Networks (GANs) for image generation.
 - Fine-tune **pre-trained models** for custom tasks using **Hugging Face Transformers** and **Vision Transformers (ViT)**.
 - Apply **image transformation techniques** such as style transfer and super-resolution.
 - Explore **trending AI/ML topics**, including generative AI, reinforcement learning, and ethical AI practices.

What will NOT be covered:

- Advanced mathematical / statistical concepts that drive the machine learning algorithms (only practical applications will be covered).
- In-depth ML features of Scikit-Learn, Keras and TensorFlow due to schedule constraints. Focus will be on PyTorch, NumPy and HuggingFace Transformers.

Training mode:

- The Training program will be mostly demonstration and discussion oriented.
- Most concepts will be taught by demonstrating live code and participants are expected to learn by practicing the same.
- Most topic/library/function coverage will be accompanied with a hands-on exercise / home-work assignment for the participants.

Lab requirements:

As the training is highly lab oriented, each participant attending the training program must be provided with a computer/laptop with the following software installed:

- Anaconda Python distribution installed and configured.
- VisualStudio Code with Python extensions enabled.

Course Agenda

Day 1: Introduction to Python for Machine Learning and Jupyter Notebook

- **Objective:** Set up the environment and introduce Python tools for ML.
- **Topics:**
 - Overview of Python for Machine Learning
 - Setting up the environment: Anaconda, Python, and Jupyter Notebook
 - Jupyter Notebook basics: cells, shortcuts, and magic commands
 - An overview on Python tools and frameworks bundled with Anaconda Python
 - Hands-on: Writing and running Python code in Jupyter Notebook
- **Assignment:** Create a Jupyter Notebook with basic Python operations.

Day 2: Data Manipulation with NumPy and Image Basics

- **Objective:** Learn numerical computing with NumPy and basics of image processing.
- **Topics:**

- Introduction to NumPy: arrays, shapes, and operations
 - Array indexing, slicing, and reshaping
 - Mathematical operations: linear algebra, statistics, and random sampling
 - Basics of image representation: pixels, color channels, and transformations
 - Hands-on: Manipulating images using NumPy (e.g., cropping, resizing, filtering)
 - **Assignment:** Perform matrix operations and apply basic image transformations.
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Day 3: Data Analysis with Pandas and Image Data Preprocessing

- **Objective:** Master data manipulation with Pandas and preprocess image data.
 - **Topics:**
 - Introduction to Pandas: Series and DataFrames
 - Data cleaning: handling missing data, duplicates, and outliers
 - Data transformation: filtering, grouping, and aggregation
 - Image data preprocessing: normalization, augmentation, and dataset preparation
 - Hands-on: Analyzing a dataset and preprocessing image data for ML
 - **Assignment:** Clean and preprocess an image dataset for training.
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Day 4: Introduction to PyTorch

- **Objective:** Learn the basics of PyTorch.
 - **Topics:**
 - Introduction to PyTorch: tensors and neural networks
 - 1D and 2D tensors
 - Differentiation in PyTorch
 - Using `autograd` and configuring GPU acceleration
 - Training and Predicting using Linear Regression
 - Hands-on: Try out simple classifier and regressions using PyTorch
 - **Assignment:** Explore PyTorch features: backpropagation, activation functions and softmax regression.
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Day 5: Image Recognition with CNNs

- **Objective:** Use PyTorch to build image recognition models.
 - **Topics:**
 - An overview on FFNN, RNN and CNN
 - Building Convolutional Neural Networks (CNNs) for image recognition
 - Training and evaluating CNN models
 - Hands-on: Building a CNN for image classification (e.g., MNIST or CIFAR-10)
 - **Assignment:** Train a CNN on a small image dataset.
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Day 6: Advanced Image Recognition and Transfer Learning

- **Objective:** Dive deeper into image recognition and transfer learning.
- **Topics:**

- Advanced CNN architectures: ResNet, VGG, and Inception
 - Transfer learning with pre-trained models (e.g., using `torchvision`)
 - Fine-tuning pre-trained models for custom datasets
 - Hands-on: Fine-tuning a pre-trained model for a custom image classification task
 - **Assignment:** Fine-tune a pre-trained model on a custom image dataset.
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Day 7: Image Generation and Transformation Techniques

- **Objective:** Explore image generation and transformation using deep learning.
 - **Topics:**
 - Introduction to Generative Adversarial Networks (GANs)
 - Image generation using GANs (e.g., DCGAN, StyleGAN)
 - Image transformation techniques: super-resolution, style transfer, and image-to-image translation
 - Hands-on: Building a simple GAN for image generation or applying style transfer
 - **Assignment:** Generate images using a GAN or apply style transfer to an image.
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Day 8: Hugging Face Transformers for NLP and Vision

- **Objective:** Explore state-of-the-art NLP and vision techniques using Hugging Face.
 - **Topics:**
 - Introduction to Hugging Face Transformers
 - Pre-trained models for text classification, summarization, and translation
 - Vision transformers (ViT) for image recognition
 - Hands-on: Fine-tuning a transformer model for NLP or vision tasks
 - **Assignment:** Fine-tune a transformer model for a text or image classification task.
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Case study project exercise

Building an Intelligent Image Captioning System

Project Overview:

Participants should build an **intelligent image captioning system** that generates descriptive captions for images. This project should combine **computer vision** (image recognition) and **natural language processing** (text generation) to create a practical and engaging application.

Project Steps:

1. **Data Collection and Preprocessing:**
 - Use a publicly available dataset like **COCO (Common Objects in Context)** or **Flickr8k**, which contains images paired with captions.
 - Preprocess the images (resize, normalize, etc.) and tokenize the captions for NLP.
2. **Image Recognition with CNNs:**
 - Use a pre-trained CNN model (e.g., ResNet or VGG) to extract features from the images.

- Fine-tune the CNN if necessary for better feature extraction.

3. Text Generation with Transformers:

- Use a pre-trained language model (e.g., GPT or a Hugging Face Transformer) to generate captions.
- Combine the image features with the text generation model to create a unified architecture.

4. Model Training:

- Train the combined model using the image-caption pairs.
- Use techniques like **teacher forcing** to improve training efficiency.

5. Evaluation:

- Evaluate the model using metrics like **BLEU score** or **CIDEr** to measure the quality of generated captions.
- Test the model on unseen images to assess its generalization ability.

6. Deployment:

- Deploy the model as a simple web application using **Flask** or **Streamlit**.
- Allow users to upload images and receive generated captions in real-time.

Tools and Frameworks:

- **PyTorch**: For building and training the CNN and transformer models.
- **Hugging Face Transformers**: For pre-trained language models.
- **NumPy/Pandas**: For data preprocessing.
- **Flask/Streamlit**: For deploying the model as a web app.